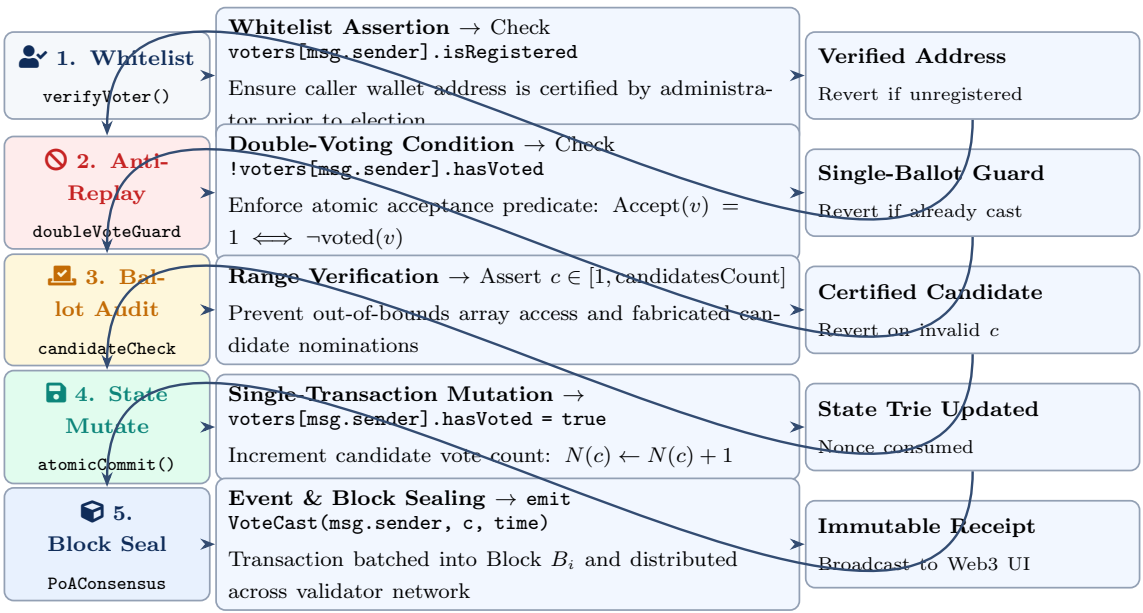

1. Whitelist
verifyVoter()
Whitelist Assertion → Check voters[msg.sender].isRegistered
Ensure caller wallet address is certified by administrator prior to election
Verified Address
Revert if unregistered
2. Anti-Replay
doubleVoteGuard
Double-Voting Condition → Check !voters[msg.sender].hasVoted
Enforce atomic acceptance predicate: Accept(v) = 1 ⟺ ¬voted(v)
Single-Ballot Guard
Revert if already cast
3. Ballot Audit
candidateCheck
Range Verification → Assert c ∈ [1, candidatesCount]
Prevent out-of-bounds array access and fabricated candidate nominations
Certified Candidate
Revert on invalid c
4. State Mutate
atomicCommit()
Single-Transaction Mutation → voters[msg.sender].hasVoted = true
Increment candidate vote count: N(c) ← N(c) + 1
State Trie Updated
Nonce consumed
5. Block Seal
PoAConsensus
Event & Block Sealing → emit VoteCast(msg.sender, c, time)
Transaction batched into Block B_i and distributed across validator network
Immutable Receipt
Broadcast to Web3 UI